Gas Mixtures and Gas Stoichiometry

Answer key with worked solutions

A. Partial pressures

1. Dalton's law.

> The total pressure of a mixture of non-reacting gases is the **sum of the partial pressures**. At fixed V and T, each gas's pressure depends only on its number of moles ($P = nRT/V$), so its share of the pressure equals its share of the moles: $P_{\text{A}} = x_{\text{A}} P_{\text{total}}$.

2. Air.

> N_2: $0.781 \times 1.00 \text{ atm} = \mathbf{0.781 \text{ atm}}$ = **79.1 kPa**
>
> O_2: $0.209 \times 1.00 \text{ atm} = \mathbf{0.209 \text{ atm}}$ = **21.2 kPa**
>
> Ar: $0.0093 \times 1.00 \text{ atm} = \mathbf{0.0093 \text{ atm}}$ = **0.94 kPa**

3. Helium and oxygen.

> (a) $n_{\text{He}} = \dfrac{2.00 \text{ g}}{4.003 \text{ g/mol}} = 0.4996 \text{ mol}$; $n_{\text{O}_2} = \dfrac{8.00 \text{ g}}{32.00 \text{ g/mol}} = 0.2500 \text{ mol}$
>
> (b) $x_{\text{He}} = \dfrac{0.4996 \text{ mol}}{0.7496 \text{ mol}} = \mathbf{0.667}$
>
> (c) $T = 27 + 273.15 = 300.15 \text{ K}$
>
> $$P_{\text{He}} = \frac{0.4996 \text{ mol} \times 0.08206 \ \frac{\text{L atm}}{\text{mol K}} \times 300.15 \text{ K}}{10.0 \text{ L}} = \mathbf{1.23 \text{ atm}}$$
>
> P_{O_2} (half the moles of He) = **0.616 atm**; $P_{\text{total}} = 1.23 \text{ atm} + 0.616 \text{ atm} = \mathbf{1.85 \text{ atm}}$

© 2026 Chemistry Clarity · chemistryclarity.com · Free for personal and classroom use. Not for resale.

B. Gases collected over water

4. Oxygen over water.

> (a) $P_{O_2} = 748\ \text{mmHg} - 19.8\ \text{mmHg} = 728.2\ \text{mmHg}$; $\dfrac{728.2\ \text{mmHg}}{760\ \text{mmHg/atm}} = \mathbf{0.9582\ atm}$
>
> (b) $T = 22 + 273.15 = 295.15\ \text{K}$
>
> $$n = \frac{PV}{RT} = \frac{0.9582\ \text{atm} \times 0.500\ \text{L}}{0.08206\ \frac{\text{L atm}}{\text{mol K}} \times 295.15\ \text{K}} = \mathbf{0.0198\ mol}$$
>
> Mass: $0.01978\ \text{mol} \times 32.00\ \text{g/mol} = \mathbf{0.633\ g}$

C. Gas stoichiometry

5. Hydrogen from zinc.

> $$n(\text{Zn}) = \frac{1.31\ \text{g}}{65.38\ \text{g/mol}} = 0.02004\ \text{mol} = n(\text{H}_2)$$
>
> $$V = \frac{nRT}{P} = \frac{0.02004\ \text{mol} \times 0.08206\ \frac{\text{L atm}}{\text{mol K}} \times 298.15\ \text{K}}{1.00\ \text{atm}} = \mathbf{0.490\ L}$$

6. Propane.

> Volume ratios equal mole ratios. (a) $2.00\ \text{L} \times 5 = \mathbf{10.0\ L\ O_2}$ (b) $2.00\ \text{L} \times 3 = \mathbf{6.00\ L\ CO_2}$

7. Molar volume.

> $0.250\ \text{mol} \times 22.4\ \text{L/mol} = \mathbf{5.60\ L}$. 22.4 L/mol applies only at 0 °C and 1 atm; at 25 °C the gas expands (24.5 L/mol), so use $PV = nRT$ instead.

© 2026 Chemistry Clarity · chemistryclarity.com · Free for personal and classroom use. Not for resale.

D. Challenge

8. Airbag.

Moles of N_2:

$$n = \frac{PV}{RT} = \frac{1.10\ \text{atm} \times 60.0\ \text{L}}{0.08206\ \frac{\text{L atm}}{\text{mol K}} \times 298.15\ \text{K}}$$
$$= 2.698\ \text{mol N}_2$$

Moles and mass of NaN_3 (2 NaN_3 : 3 N_2):

$$2.698\ \cancel{\text{mol N}_2} \times \frac{2\ \cancel{\text{mol NaN}_3}}{3\ \cancel{\text{mol N}_2}} \times \frac{65.02\ \text{g}}{1\ \cancel{\text{mol NaN}_3}}$$
$$= \mathbf{117\ g\ NaN_3}$$

© 2026 Chemistry Clarity · chemistryclarity.com · Free for personal and classroom use. Not for resale.